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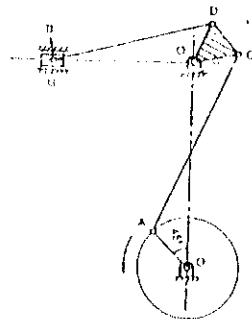
B.Tech. Degree V Semester Examination, November 2005

ME 501 MECHANICS OF MACHINERY
(2002 Admissions)

Time: 3 Hours

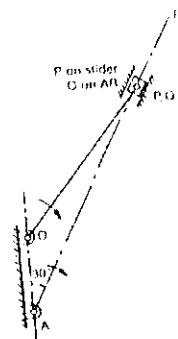
Maximum Marks: 100

- I
- Derive an expression for the travel of ram in a 'Crank and Slotted Lever' Quick Return Motion Mechanism. (6)
 - The lengths of various links of a mechanism shown in figure are as follows: OA=150mm, AC=600mm, CQ=QD=145mm, CD=125mm, BD=500mm and OQ=625mm. The crank OA rotates at 60 rpm in the counter clockwise direction. Determine the velocity of the slider B and the angular velocity of the link BD when the crank has turned an angle of 45° with the vertical. (14)



OR

- II
- Derive an expression for the magnitude of Coriolis component of acceleration. (6)
 - One cylinder of a rotary engine is shown in the configuration diagram as shown in the figure. OA is the fixed crank, 200mm long. OP is the connecting rod and is 520mm long. The line of stroke is along AR and at the instant is inclined at 30° to the vertical. The body of the engine consisting of cylinders rotate at a uniform speed of 400 rpm about the fixed center A. Determine (i) acceleration of piston inside the cylinder and (ii) angular acceleration of the connecting rod. (14)



- III
- Determine the maximum and minimum transmission angles for a slider crank mechanism. (5)
 - Explain the method of obtaining the co-ordinates of a coupler point in a four bar mechanism. (5)
 - Explain the Overlay method in the synthesis of a function generator. (10)

OR

- IV
- Obtain the Freudenstein's equation for a four bar mechanism. (6)
 - Determine the crank and coupler lengths of a four bar mechanism to satisfy the conditions : $\omega_{AP}=15 \text{ rad/s}$, $\omega_{OP}=10 \text{ rad/s}$, $\sigma_{AP}=0 \text{ rad/s}^2$ and $\alpha_{OP}=15 \text{ rad/s}^2$ at $\phi = 45^\circ$ (10)

(Turn Over)

when the lengths of the fixed link and the follower are 5cm and 6cm respectively. ϕ is the follower angle. ω_{ip} and ω_{op} are the angular velocities for the input and output link. α_{ip} and α_{op} are the angular accelerations for the input and output link. (14)

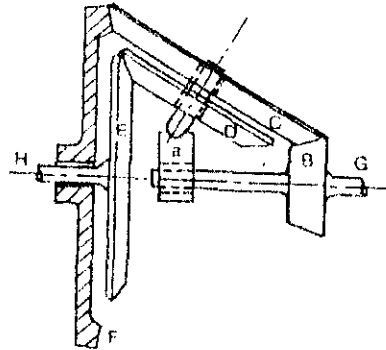
- V a) Explain with neat sketch, the working of an intermittent motion mechanism. (6)
 b) The following data relate to a cam operating an oscillating roller follower. Minimum diameter of the cam = 44mm, Diameter of roller = 14mm, Length of follower arm = 40mm, Distance of fulcrum center from cam center = 50mm, Angle of ascent = 75° , Angle of descent = 105° , Angle of dwell for the follower in the highest position = 60° , Angle of oscillation of follower = 28° , Draw the profile of the cam if the ascent and descent both take place with SHM. (14)

OR

- VI a) Explain any one type of approximate straight line motion mechanism with the help of a sketch. (6)
 b) A tangent cam with a base circle diameter of 50mm operates a roller follower 20mm in diameter. The line of stroke of the roller follower passes through the axis of the cam. The angle between the tangential faces of the cam is 60° , speed of cam shaft 200 rpm and the lift of the follower 15mm, find the main dimensions of the cam. Also determine the acceleration of the follower at the beginning of lift. (14)
- VII a) Derive an expression for the length of path of contact in a meshing gear. (6)
 b) A pair of 20° full depth involute spur gears having 30 and 50 teeth respectively of module 4mm are in mesh. The smaller gear rotates at 1000 rpm. Determine (i) sliding velocity at engagement and at disengagement of a pair of teeth and (ii) contact ratio. (14)

OR

- VIII a) Derive an expression for velocity of sliding between a pair of teeth in meshing gears. (6)
 b) The figure shows a Humpage gear train used in a lathe headstock. The number of teeth on the wheels B, C, D, E and F are 18, 54, 22, 44 and 72 respectively. If the shaft G rotates at 200 rpm clockwise, determine the speed of shaft H when F is rotated at 300 rpm opposite to G. (14)



- IX a) Explain with neat sketch, the working of any one type of dynamometer. (6)
 b) Power is transmitted using a V-belt drive. The included angle of V-groove is 30° . The cross sectional area of the belt is 400mm^2 . If the mass of the belt is 0.35 Kg per metre length and the maximum allowable stress is 1.4 M.Pa, determine the power transmitted if the condition of maximum power prevails. The angle of lap is 140° and $\mu = 0.15$. (14)

OR

- X a) Derive a relation for the maximum torque transmitted by an old single plate clutch of external and internal radii 'R' and 'r', if the axial spring load is 'W'. (6)
 b) A band brake acts on the $3/4^{\text{th}}$ circumference of a drum of 450mm diameter, which is keyed to the shaft. The band brake provides a braking torque of 225 Nm. One end of the band is attached to a fulcrum pin of the lever and the other end to a pin 100mm from the fulcrum. If the operating force is applied at 500mm from the fulcrum and the coefficient of friction is 0.25, determine the operating force when the drum rotates in the clockwise direction. (14)
